

An Efficient IoT Device Classification Method Based on the Integration of Variational Autoencoder and XGBoost

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Abstract

This paper proposes an effective IoT device classification method combining Variational Autoencoder (VAE) and an optimized XGBoost model to address challenges like large-scale data processing and sample imbalance. First, feature selection is improved by integrating mutual information analysis and domain knowledge. VAE is used for feature extraction, capturing latent representations from network traffic data. The XGBoost classifier is optimized using techniques like gradient clipping and model pruning to improve training efficiency and reduce computational overhead. This method boosts classification accuracy, handling data imbalance and noise effectively, with experimental results showing 90% accuracy and F1 score. The proposed approach outperforms traditional algorithms in large-scale IoT device classification tasks.

CCS Concepts

• **Computer systems organization** → **Embedded systems**; *Redundancy*; Robotics; • **Networks** → Network reliability.

Keywords

Internet of Things, Network Traffic, Neural Networks, Machine Learning

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1 Introduction

The rapid growth of IoT devices has advanced areas like smart homes, healthcare, and industrial automation, but also introduced significant security risks. Automated IoT device identification through network traffic is a key solution, enabling security policy enforcement and management actions without relying on manufacturer infrastructure [1].

Current techniques for IoT device identification focus on network traffic analysis, typically captured by central nodes like routers. This approach ensures privacy and doesn't depend on cloud services, as IoT devices' network behavior accurately reflects their type. However, existing methods suffer from accuracy degradation when adapting to new datasets or devices, requiring frequent retraining.

Traditional machine learning methods, such as SVM, decision trees, and random forests, extract handcrafted features from network traffic for device identification. While effective in simple scenarios, these methods rely on manually designed features and struggle with complex network patterns and large-scale datasets, limiting their adaptability [2]. As IoT devices diversify and network environments change, traditional methods face growing limitations [3].

Deep learning techniques have been increasingly adopted for IoT device classification, offering advantages in capturing complex patterns and maintaining high classification performance, particularly in dynamic environments and with many unknown devices [4]. However, challenges remain, especially with data heterogeneity and device behavior diversity. To summarize, identifying IoT devices through network traffic continues to encounter various challenges in real-world applications:

- (1) Data heterogeneity and diversity: IoT devices are diverse, and network traffic behavior patterns are complex and variable, leading to a decline in classification accuracy when adapting to new devices and dynamic environments using both traditional and deep learning methods.
- (2) Computational resources and model complexity: Although deep learning models can capture more complex features,

their high computational and memory overhead in large-scale IoT network traffic is still a bottleneck, especially in resource-constrained environments.

- (3) Real-time and online learning requirements: The dynamic updates and continuous changes in IoT devices require the recognition system to be real-time and capable of adapting to new devices in uninterrupted network traffic, maintaining efficient performance.

To address these challenges, this paper proposes an efficient classification method based on the integration of Variational Autoencoder (VAE) and XGBoost. The method first performs feature selection based on mutual information and domain knowledge, and then uses VAE for feature extraction, discovering latent feature representations from network traffic data via unsupervised learning. It then employs the XGBoost classifier to efficiently classify the extracted features. By deeply combining VAE and XGBoost, the proposed method can achieve more accurate and robust device classification in complex IoT environments while effectively reducing computational resource consumption.

The main contributions of this paper are as follows:

- (1) We design an IoT device classification framework based on the integration of VAE and XGBoost, fully leveraging VAE's unsupervised feature learning and the powerful classification capability of XGBoost to enhance classification accuracy.
- (2) We evaluate the method on large-scale network traffic datasets, demonstrating its superior performance in the classification of various IoT devices, especially in addressing the data imbalance problem.
- (3) We propose a feature selection method based on mutual information and domain knowledge, effectively selecting a feature set highly relevant to IoT device recognition.
- (4) We provide new insights into the application potential of deep learning models in IoT device recognition, offering guidance for future research and practical applications.

The structure of this paper is as follows: Section 2 reviews the related literature; Section 3 outlines the IoT device classification framework that combines VAE and XGBoost; Section 4 presents the experimental results for the proposed method; Section 5 summarizes the findings and explores future research directions.

2 RELATED WORK

IoT device identification is essential for network security and device management. Research focuses on network traffic-based classification methods using traditional machine learning and deep learning techniques. Traditional methods like SVM and Random Forest extract features, while deep learning methods like CNN and LSTM enhance accuracy, particularly in dynamic environments. However, challenges remain with generalization and adaptability, especially with diverse IoT devices and fluctuating network conditions [5]. Traffic analysis is central to device identification, with studies focusing on feature extraction and classification. Desai et al. proposed a framework ranking features using statistical methods for classification [7], and Meidan et al. introduced a multi-stage machine learning classifier [8]. These methods struggle with encrypted traffic and lack scalability in dynamic environments. Traditional methods rely on unique identifiers like MAC addresses, but issues

like hidden or spoofed identifiers persist. Behavior-based recognition has gained traction but still struggles with adaptability for unknown devices and dynamic environments [9]. Traditional machine learning techniques like SVM, KNN, and Random Forests are effective in resource-constrained environments [1][10]. Cvitić et al. proposed an ensemble machine learning model for smart home IoT device classification, but it needs optimization for more complex environments [11]. Jmila et al. explored machine learning-based network traffic analysis for smart home device classification, showing improved accuracy but lacking real-time adaptability and diverse datasets [12]. Deep learning methods such as 1D-CNN, LSTM, and attention mechanisms are effective in recognizing IoT devices from raw traffic data [13]. Bai et al. introduced an LSTM-CNN cascading model, but its accuracy decreases with more device categories and small datasets [14]. Bao et al. proposed a hybrid deep learning framework that balances accuracy and computational load, but it faces challenges with real-time performance [15]. Despite advancements, existing methods struggle with dynamic device behavior, and deep learning methods are computationally expensive for resource-limited devices [16]. This paper proposes a classification method integrating Variational Autoencoders (VAE) and XGBoost, enhancing adaptability, accuracy, and robustness for IoT device recognition.

3 DESIGN

3.1 Model overview

The proposed model for IoT device classification combines Variational Autoencoder (VAE) and XGBoost to handle challenges such as data heterogeneity, sample imbalance, and computational efficiency. As illustrated in Figure 1, the model begins with VAE for unsupervised feature extraction from network traffic data, capturing the latent features. These extracted features are then passed to the XGBoost classifier for classification. The process is outlined in Algorithm 1, which details the steps for feature selection, data normalization, VAE training, feature encoding, and final classification with XGBoost. This integrated approach allows the model to benefit from the unsupervised feature learning of VAE and the robust classification power of XGBoost, improving classification accuracy while reducing computational resources.

3.2 Feature selection

This study presents a feature selection method for IoT device identification that combines mutual information analysis with domain knowledge. Mutual information quantifies the dependency between features and the target variable, capturing both linear and nonlinear correlations [17-18]. Domain knowledge helps optimize these results, preventing the exclusion of features that may have low mutual information but are important in practice.

The dataset includes various traffic statistics and protocol features, such as outgoing and incoming bytes ($bytes_{out}$, $bytes_{in}$), packet counts ($num_{pktsout}$, num_{pktsin}), time-related features ($f_{iptmean}$, f_{iptvar}), and payload-related features (f_bmean , f_bstd). Mutual information is used to calculate the contribution of each feature to the target variable, enabling the removal of irrelevant or redundant features.

However, mutual information alone may overlook features with weak associations to the target variable. Domain knowledge indicates that $bytes_{out}$, $num_{pktsout}$, and $f_{iptmean}$ are crucial for describing traffic behavior, even with low mutual information scores. Features like $f_{iptmean}$ and f_{iptvar} combine to capture temporal traffic patterns critical for identifying IoT devices.

The final feature selection process involves combining mutual information scores with domain knowledge, assigning higher weights to key features and lower weights to less relevant ones. Cross-validation results show that this optimized feature selection improves the model's performance.

Key features prioritized for retention include:

Traffic statistics: $bytes_{out}$, $num_{pktsout}$, $bytes_{in}$, num_{pktsin} ;

Time series features: $f_{iptmean}$, f_{iptvar} ;

Payload distribution features: f_bmean , f_bstd .

By combining mutual information and domain knowledge, the method ensures a balanced selection of features, enhancing model interpretability, efficiency, and performance.

3.3 Data preprocessing

In the study, to meet the deep learning model's requirements for modeling time series features, this paper introduces the operation of serializing two-dimensional feature sequences into three-dimensional inputs during the data preprocessing stage to ensure the effective representation of data in the deep learning model. The specific preprocessing process is as follows: Firstly, in the original dataset, there may be significant differences in the value range and scale of feature dimensions. To eliminate the impact of different feature units, the Min-Max normalization method is used to normalize the features, with the following formula:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (1)$$

where X is the original feature value; X_{min} and X_{max} are the minimum and maximum values of the corresponding feature; X_{norm} is the normalized feature value, with all values mapped to the $[0, 1]$ rang. After standardization, the original data has the following feature structure: the data shape is a two-dimensional matrix, denoted as $X \in \mathbb{R}^{n \times m}$, where n is the number of samples, and m is the number of features, meaning each sample has m features.

To adapt to the input requirements of deep learning model, the aforementioned two-dimensional feature matrix is converted into a three-dimensional input tensor. The specific conversion method is as follows: a time-step dimension is introduced, treating the feature dimension $m = 19$ as the time steps, assuming each time step corresponds to a feature value. The converted data shape is $X' \in \mathbb{R}^{n \times t \times f}$, where:

- t is the time step length, equal to the original number of features m ;
- f is the number of features per time step, set to 1 (i.e., each time step has only one feature).

Accordingly, the original two-dimensional data $X \in \mathbb{R}^{n \times 19}$ is reshaped into three-dimensional data $X' \in \mathbb{R}^{n \times 19 \times 1}$.

The converted data structure treats each sample as a sequence, containing 19 time steps, with one feature for each time step. This three-dimensional representation allows the model to capture the local relationships or sequential dependencies between features

through the time dimension (timesteps), which is particularly suitable for convolution operations and recurrent neural networks. The preprocessing step of transforming the data from two dimensions to three dimensions is one of the crucial components in the modeling approach of this paper. Through this operation, the original feature sequence is reorganized into an information structure with a time dimension, providing a more suitable input format for the subsequent deep learning model construction.

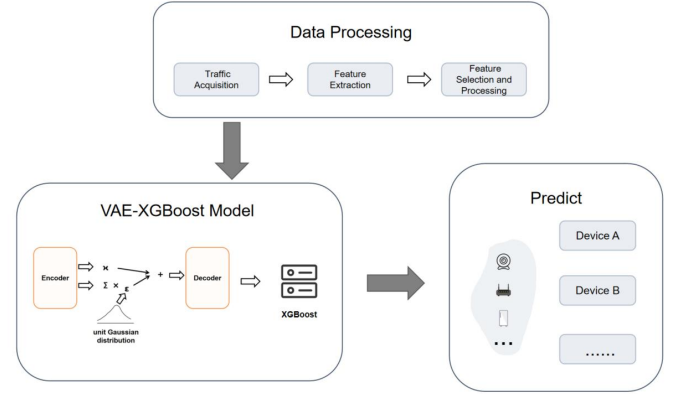


Figure 1: This is the framework diagram of the proposed model.

3.4 Variational Autoencoder for Feature Extraction

To obtain latent feature representations from IoT device network traffic data, we present an innovative unsupervised learning approach based on the VAE. Unlike conventional autoencoders, VAE utilizes a probabilistic model that learns the underlying distribution of the data by optimizing the Evidence Lower Bound (ELBO). This design not only generates smooth and continuous latent representations but also effectively captures intricate traffic patterns, particularly when handling the diversity of IoT devices and the constantly changing network conditions. By incorporating an adaptive reconstruction mechanism and multi-scale feature learning, we improve the model's generalization capabilities and computational efficiency in handling large and high-dimensional datasets.

As shown in Figure 2: VAE Framework Diagram, the VAE model structure consists of two parts: the encoder and the decoder. The encoder compresses the input network traffic data into a representation in the latent space, which is not only a low-dimensional representation of the data but also embeds the probabilistic distribution characteristics of the data. Unlike traditional autoencoders, VAE models the latent space probabilistically, learning the underlying distribution of the data and using reparameterization to ensure smoothness and continuity of the latent space. This allows the model to effectively perform data generation and feature extraction. In Figure 2, we can see the following components and parameters involved in the VAE model:

As shown in Figure 2. VAE Framework Diagram, the VAE model structure consists of two parts: the **encoder** and the **decoder**.

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In the encoder design, the input data is initially processed through a fully connected layer containing 128 neurons and activated by the ReLU function. It then passes through another fully connected layer with 64 neurons, where it is mapped to the mean (z_{mean}) and logarithmic variance (z_{logvar}) of the latent space. This configuration ensures that the VAE effectively captures the statistical properties of the latent space and is adaptable to the complexities of IoT device data. To guarantee that the latent space follows a standard normal distribution, VAE employs a sampling method and a Lambda layer for reparameterization, enhancing the quality and control over the latent space.

The decoder takes the latent variable (z) as input. After passing through two fully connected layers with 64 and 128 neurons, respectively, it produces an output that maintains the same dimensionality as the original input. To ensure the output data aligns with the actual network traffic range, the decoder applies the sigmoid activation function, which restricts the output values to the $[0, 1]$ range.

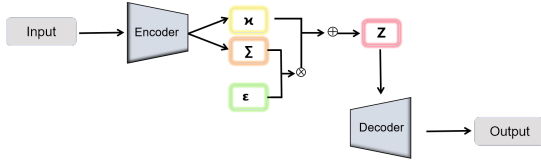


Figure 2: VAE Framework Diagram

The VAE loss function consists of two components: the reconstruction loss and the KL divergence loss. The reconstruction loss calculates the difference between the decoded output and the original input data using MSE, guiding the model to reconstruct an output that closely resembles the input. The KL divergence loss quantifies the difference between the latent distribution produced by the encoder and the standard normal distribution, assisting in optimizing the latent space distribution. The calculation for the KL divergence loss is as follows:

$$KL\ Loss = -0.5 \times \sum (1 + z_{logvar} - z_{mean}^2 - \exp(z_{logvar})) \quad (2)$$

Additionally, this paper introduces an adaptive reconstruction loss mechanism, where the reconstruction loss weight is dynamically adjusted based on the traffic features of different device types. This allows the model to optimize its learning ability for different features, depending on the traffic complexity and diversity of the devices, further enhancing the model's performance and stability.

The overall VAE loss function is the combination of the reconstruction loss and the KL divergence loss, where both loss components are optimized to achieve the most accurate latent space representation and reconstruction outcome.

$$VAELoss = \text{Reconstruction Loss} + KLLoss \quad (3)$$

With these advancements, VAE effectively captures relevant latent features from the complex network traffic data of IoT devices, addressing the challenges typically encountered in managing high-dimensional and large-scale data. Particularly, by incorporating a multi-scale feature learning approach, VAE extracts features at various levels, improving the model's ability to adapt to different IoT devices and network conditions. This innovation enhances both the classification accuracy and computational efficiency, reducing resource consumption and making the method more scalable and effective in handling large datasets and complex application scenarios.

By combining the strengths of VAE and XGBoost, the proposed IoT device classification method efficiently extracts features and performs accurate classification, while addressing issues such as data imbalance and excessive computational resource consumption. This demonstrates its great application potential in the field of IoT device recognition.

Algorithm 1 IoT Device Classification Using VAE and XGBoost

Input: IoT Device Network Traffic Data, Feature Set F , Preprocessed Traffic Data, Hyperparameters for VAE and XGBoost.

Output: Predicted IoT Device Categories.

- 1: Initialize an empty list for selected features: $List_f \leftarrow []$
 - 2: // **Perform feature selection based on mutual information and domain knowledge**
 - 3: $SelectedFeatures \leftarrow \text{SELECTFEATURES}(F)$
 - 4: $List_f.append(SelectedFeatures)$
 - 5: Normalize the traffic data
 - 6: $TrafficDataNorm \leftarrow \text{NORMALIZEDATA}(SelectedFeatures)$
 - 7: Reshape the data for model input
 - 8: $ReshapedData \leftarrow \text{RESHAPE}(TrafficDataNorm)$
 - 9: // **Train the VAE model for feature extraction**
 - 10: $VAEModel \leftarrow \text{TRAINVAE}(ReshapedData)$
 - 11: $EncodedFeatures \leftarrow \text{ENCODE}(VAEModel, ReshapedData)$
 - 12: // **Implement adaptive reconstruction loss mechanism to adjust weights based on traffic features of different devices**
 - 13: $EncodedFeatures \leftarrow \text{ADAPTCONSTRUCTION}(EncodedFeatures)$
 - 14: // **Train the XGBoost classifier using the encoded features with gradient clipping and pruning**
 - 15: $XGBoostModel \leftarrow \text{TRAINXGBOOSTWITHOPT}(EncodedFeatures)$
 - 16: // **Predict IoT device categories using the trained XGBoost model**
 - 17: $PredictedCategories \leftarrow \text{PREDICT}(XGBoostModel, EncodedFeatures)$
 - 18: **Return** $PredictedCategories$
-

3.5 XGBoost Classifier

In the feature extraction phase, latent space representations from VAE are input to the XGBoost classifier, which uses these features to classify IoT device categories. To improve computational efficiency, XGBoost incorporates gradient clipping and model pruning, reducing memory consumption and mitigating overfitting, particularly in high-dimensional feature spaces.

XGBoost utilizes Gradient Boosting, iteratively training weak classifiers (decision trees) to improve classification performance. Gradient clipping controls extreme gradients to prevent instability, while model pruning eliminates unnecessary branches, reducing computation. Key hyperparameters include:

n_estimators: 100 trees,
learning_rate: 0.1,
max_depth: 6,
 objective: 'multi:softmax' for multi-class tasks,
num_class: number of IoT device types.

XGBoost's built-in feature selection automatically identifies the most important features for classification. The VAE-extracted features further optimize this process, allowing XGBoost to efficiently assess feature contributions, improving classification accuracy.

Through gradient clipping and model pruning, XGBoost maintains high accuracy while reducing computational costs, offering an efficient solution for large-scale IoT device data.

4 EXPERIMENTAL EVALUATION

4.1 Experiment Setup

The experiment is conducted using an NVIDIA GeForce RTX 3080 GPU with 10 GB of memory and two Intel(R) Xeon(R) Silver 4210R CPUs @ 2.40GHz with 20 cores. After performing data standardization and reshaping it into a 3D format, the dataset is split into a training set and a test set.

4.2 Dataset

The dataset used in this study was sourced from [1] and collected through a laboratory testbed, containing data from 41 IoT devices across domains like smart homes, environmental monitoring, and security. Data was gathered using a Linux server with a Wi-Fi card supporting 2.4 GHz and 5 GHz frequency bands, and a 1 Gbps Ethernet interface. IoT device traffic was captured via tcpdump and stored as pcap files, categorized by device MAC address. Data collection occurred over 27 weeks with sparse interactions, mainly triggered by automatic activations or occasional user operations. The dataset was divided into three 9-week periods for training and evaluation of the models.

4.3 Experimental Results

In this experiment, we compare the performance of several models for IoT device classification, as shown in Table 1. The models are as follows: (1) IoT-ClassNN[1]: This model is based on a simple neural network architecture designed for classifying IoT devices. It performs poorly in both Week 1 and Week 2, with low accuracy and F1 scores. (2) IoT-CK[15]: This model uses a clustering-based approach with KMeans for IoT device classification. It performs

Table 1: Comparison Experiment Results with Several Models

Model	Week 1			Week 2		
	Accuracy	Recall	F1	Accuracy	Recall	F1
IoT-ClassNN	0.26	0.26	0.26	0.25	0.26	0.27
IoT-CK	0.50	0.33	0.32	0.50	0.35	0.33
IoT-LC	0.55	0.44	0.44	0.64	0.53	0.53
Ours	0.89	0.90	0.90	0.89	0.89	0.90

moderately in Week 1 with an accuracy of 0.50 and shows improvement in Week 2, reaching an accuracy of 0.64, but still does not outperform the Ours model. (3) IoT-LC[14]: This model combines LSTM and CNN to capture the time-series characteristics of IoT device network traffic. It shows an improvement in Week 2 with an accuracy of 0.64 and an F1 score of 0.53. Finally, the Ours model, which integrates VAE (Variational Autoencoder) and XGBoost, outperforms all these models in both weeks, achieving high accuracy, recall, and F1 scores of 0.89 in Week 1 and 0.89 in Week 2.

Table 2: Comparison Experiment Results with Standard Classifiers

Model	Week 1			Week 2		
	Accuracy	Recall	F1	Accuracy	Recall	F1
RF	0.77	0.72	0.73	0.72	0.72	0.72
DT	0.83	0.82	0.82	0.81	0.80	0.80
SVM	0.80	0.79	0.79	0.79	0.73	0.75
Ours	0.89	0.90	0.90	0.89	0.89	0.90

In this experiment, as shown in Table 2, the Ours model, which combines the Variational Autoencoder (VAE) with XGBoost, performs excellently in the IoT device identification task, demonstrating the highest accuracy (0.89), recall (0.90), and F1 score (0.90) on both the Week1 and Week2 test sets. The Ours model effectively captures complex network traffic data patterns and outperforms traditional machine learning models in this task. In contrast, traditional machine learning models such as RF [23], DT [24], and SVM [25] perform relatively poorly. Among them, the RF model performs the worst, with lower metrics than the Ours model across all categories. Although DT and SVM perform slightly better on Week1, they show a decline in Week2, indicating weaker adaptability to different data distributions. Overall, the Ours model exhibits more stable performance across both test sets, suggesting its ability to effectively handle complex network traffic data and provide higher classification accuracy. To summarize, our model surpasses others in both accuracy and robustness, highlighting the effectiveness of deep learning techniques in IoT device identification.

In this experiment, as shown in Table 3, the Ours model underwent an ablation study on the Week 1 test set. The experimental results indicate that after removing the VAE module, the model's accuracy, recall, and F1 score all decreased, with values of 0.7, 0.66, and 0.7, respectively. After removing the XGBoost module, both accuracy and F1 score also dropped, although recall only slightly decreased. Overall, the complete Ours model (including both VAE

Table 3: Ablation Study

Model	Accuracy	Recall	F1
-VAE	0.70	0.66	0.70
-XGBoost	0.77	0.64	0.74
ALL	0.89	0.90	0.90

and XGBoost modules) performed best with accuracy (0.89), recall (0.90), and F1 score (0.90), demonstrating the superiority of the model in feature extraction and classification tasks. In conclusion, our model significantly outperforms other models in accuracy and stability, highlighting the benefits of integrating deep learning with traditional machine learning approaches, especially for complex IoT device identification tasks.

5 DISCUSSION AND LIMITATIONS

This study proposes an IoT device classification model based on network traffic, offering non-intrusive, scalable, and accurate device identification. It supports IoT device management and security, efficiently identifying devices when they first connect or when dealing with unknown devices. However, the model struggles with feature extraction in encrypted traffic environments, especially with TLS/SSL, which may decrease accuracy. It also faces challenges in dynamic network environments, handling traffic feature variations caused by user interactions, network load, or delays, while requiring high real-time performance [26][27]. The dataset's diversity and scale limitations hinder the model's generalization, as it mainly comes from a laboratory environment, not fully reflecting real-world IoT networks. Additionally, similarities in traffic features between certain devices (e.g., cameras and sensors) increase classification difficulty [22]. Future improvements could include encrypted traffic analysis, online learning, larger and more diverse datasets, and optimized feature selection methods.

6 CONCLUSION

This study presents a classification model for IoT devices combining Variational Autoencoder (VAE) and XGBoost. The model uses network traffic features and deep learning techniques to efficiently identify IoT device types, achieving high accuracy and strong scalability. It is minimally invasive, making it suitable for IoT device management and security. However, the model struggles with encrypted traffic and dynamic network environments, requiring a large, diverse dataset. Future work could improve feature selection, expand the dataset, and incorporate dynamic adaptation to enhance robustness and adaptability. This research provides valuable insights for IoT management and network security.

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